20. Solution

$\text{rate} = k[x]^n[y]^m$

$0.15 = k[0.2]^n[0.02]^m$ ——(1)

$0.30 = k[0.4]^n[0.02]^m$ ——(2)

$1.20 = k[0.4]^n[0.08]^m$ ——(3)

$(3)/(2)$

$$\frac{1.2}{0.3} = \frac{k[0.4]^n[0.08]^m}{k[0.4]^n[0.02]^m}$$

$$4 = \left(\frac{[0.08]}{[0.02]}\right)^m$$

$4 = (4)^m$

$\therefore\ m = 1$

$(2)/(1)$

$$\frac{0.30}{0.15} = \frac{k[0.4]^n[0.02]^m}{k[0.2]^n[0.02]^m}$$

$$2 = \left(\frac{[0.4]}{[0.2]}\right)^n$$

$2 = (2)^n$

$\therefore\ n = 1$

$\text{Rate} = k[x]^1[y]^1$

$0.15 = k[0.2]^1[0.02]^1$

$$\frac{0.15}{[0.2]^1[0.02]^1} = k$$

$k = 37.5\ \text{mol}^{-1}\text{L}\ s^{-1}$

23**Solution:**

We know that, $t_{1/2} = 0.693/k$

$t_{1/2} = 0.693/1.54 \times 10^{-3}\ s^{-1}$ = **450 s**

24.**Solution:**

We know that, $k = 0.693/t_{1/2}$

$k = 0.693/8.0\ \text{minutes} = 0.087\ \text{minutes}^{-1}$

For a first order reaction,

$$t = \frac{2.303}{k}\log\frac{[A_0]}{[A]}$$

$$t = \frac{2.303}{0.087\ \text{min}^{-1}}\log\left(\frac{100}{1}\right)$$

$t = 52.93\ \text{min}$

25 **Solution:**

i) Order of a reaction = 1; $t_{1/2}$ = 60 ; seconds, k = ?

We know that, $k = \frac{0.6932}{t_{1/2}}$

$$k = \frac{2.303}{60} = 0.01155\ s^{-1}$$

ii) $[A_0] = 100\%$ t = 180 s , k = 0.01155 seconds^{-1}, $[A]$ = ?

For the first order reaction

$$k = \frac{2.303}{t}\log\frac{[A_0]}{[A]}$$

$$0.01155 = \frac{2.303}{180}\log\left(\frac{100}{[A]}\right)$$

$$\frac{0.01155 \times 180}{2.303} = \log\left(\frac{100}{[A]}\right)$$

$0.9207 = \log 100 - \log[A]$

$\log[A] = \log 100 - 0.9207$

$\log[A] = 2 - 0.9207$

$\log[A] = 1.0973$

$[A] = \text{antilog of } (1.0973)$

$[A] = 12.5\%$

26 **Solution:**

i) Let A = 100M, $[A_0] - [A]$ = 20M,

For the zero order reaction

$$k = \left(\frac{[A_0] - [A]}{t}\right)$$

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